Knowledge Graph Generation from Urban Planning Texts: Climate Change Adaptation in Master Plans of New York and Shanghai

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Abstract

This study examines urban master plans from New York City and Shanghai, two representative cities in the United States and China, respectively, to explore their approaches to climate change adaptation and urban planning through knowledge graph (KG) generation. Five main aspects of great concern in climate change adaptation and five in urban planning were identified first as the nodes of the knowledge graph. Then, different techniques were applied to the two master plans to compare traditional information retrieval and new large language model (LLM) methods in generating edges connecting those concept nodes. The KGs generated are then validated against human evaluation. The results suggest that while the direct LLM method can achieve better semantic results, it lacks explainability critical for practice. A new LLM embedding-based method is proposed to combine domain expert knowledge, traditional explainable methods, and LLM embeddings to provide a more interpretable and accurate representation of concepts as KGs. The KGs generated from the two cities with different contexts also suggested the impact of different planning systems on how master plans are organized and written. The findings will be helpful to generate KGs to represent concepts and their relationships in master plans, which embed various planning theories and practice consideration in different social, economic, and regulatory contexts. The comparison will provide the basis for planning knowledge and policy transfer across different cities and cultures.

Introduction

In urban planning practice, planners often need to learn from other cities' practice as a reference in developing a new master plan. However, they usually face two Backdrop：planners can often struggles: finding time to find time to read and study plans because most are lengthy technical documents, usually hundreds of pages long, and understanding the difference of the concepts and structures in different master plans, especially those for cities in totally different contexts.. In recent years, studies have emerged to apply natural language processing (NLP) techniques to read plans. However, most of them focused on specific aspects of evaluation, with little consideration of the key concepts and their relationships embedded in the planning texts. Also, they mostly evaluated specific plans, without a comprehensive comparison of the evaluation results across cities, especially in a global context.

This study aims to fill this research gap by applying different methods in knowledge graph generation to understand the climate change adaptation in master plans in two international cities: New York and Shanghai. planners need to read an increasing number of new types of plans and thus are struggling to keep pace with international plan development. For example, for emerging planning topics such as climate change, there are hundreds of relevant plans internationally for planners to study to learn how to develop their own plans. Against this backdrop, there is an increasingly urgent need for many planners to efficiently read and process large volumes of textual information from new plans to prepare them for new challenges. Specifically, the two plans are New York City's OneNYC 2050: Building a Strong and Fair City and Shanghai's Urban Master Plan (2017–2035). were analyzed using a multi-method approach to extract, compare, and visualize key concepts.

Keywords were first identified manually from 40 climate adaptation research articles and urban planning documents, which were then analyzed supplemented by computational techniques such as word frequency analysis, TF-IDF scoring, and the Apriori algorithm to evaluate term significance and relationships. A novel LLM embedding-based method was also introduced, combining semantic similarity and entity convergence to enhance knowledge representation. Additionally, expert rankings were employed to validate and refine the results.

The extracted data were visualized as a domain knowledge graph using NetworkX, with nodes and edges representing the significance of keywords and their relationships. The comparison revealed distinctive thematic emphases: New York prioritized resilience and equity, while Shanghai focused on sustainable development and balancing urbanization with environmental protection. Evaluation metrics, including precision, recall, and divergence, were used to assess the effectiveness of the methods and highlight variations between the two cities.

This study contributes a scalable framework for analyzing urban master plans, offering insights into global urban adaptation strategies. Future research could extend this methodology to additional cities and track evolving priorities over time, enhancing its applicability in comparative urban planning studies.

Previous studies

Planning text analysis in LLM

Recently, research has increasingly focused on the evaluation and analysis of complex planning texts. Given the intricate nature of these documents, urban planners often invest substantial time in reading and extracting relevant information. In the article Can ChatGPT Evaluate Plans, Professor For example, Fu et al (2024)Xinyu introduced several methodologies that utilize ChatGPT to enhance the efficiency of planning text evaluation, including the application of TF-IDF.

Knowledge Graph Construction in urban planning Domain

Domain-specific knowledge graphs are constructed based on data relevant to a specific field. Research on domain-specific knowledge graphs has emerged to address the limitations of general knowledge graphs, which often fail to fully meet specific requirements. Some typical domain-specific knowledge graphs have been developed, such as the Geographic Information Knowledge Graph, the Medical Knowledge Graph, and the E-commerce Knowledge Graph. However, there remains a notable gap in the development of domain-specific knowledge graphs tailored for urban planning texts.

Traditional methods for constructing knowledge graphs vary depending on the requirements. For example, one approach involves directly calculating word or phrase frequencies. However, this method struggles to identify related terms effectively. Algorithms like TF-IDF are helpful for exploring topics, and cosine similarity can assist in analyzing semantic relationships between terms. Additionally, algorithms such as the Apriori algorithm can determine the importance of terms based on frequency, while traditional topic modeling can reveal relationships between themes and those relevant to our study.

With the advent of large language models (LLMs), knowledge graph construction may benefit from integrating traditional and modern techniques. For instance, Professor Xinyu Fu et al (2024) mentioned translating public demands in Deciphering Public Voice. Research shows that ChatGPT demonstrates promising results in topic modeling; however, these results remain somewhat interpretatively weak. Nonetheless, employing LLMs for modeling appears to be a feasible approach for advancing knowledge graph construction in urban planning.

Data and Methodology

Research Context

In this study, we use two master plans in New York City and Shanghai, both of those are representative and well-developed cities in U.S. and China. New York was the first city in the United States to implement zoning regulations, whereas Shanghai, as a well-developed megacity and financial hub in China, serves as a representative case study. The master paper "OneNYC 2050: BUILDING A STRONG AND FAIR CITY"" from Sustainability Agency, Mayor's Office of (Sustainability) and "Shanghai Urban Master Plan (2017–2035)" " from Shanghai Municipal People's Government.

The 'OneNYC 2050' plan emphasizes sustainability and equity, reflecting New York City's commitment to climate adaptation in a socio-economic context. On the other hand, Shanghai's Master Plan (2017–2035) focuses on balancing rapid urbanization with environmental conservation, making it a compelling comparison to explore divergent urban planning approaches.

Manual Entities identification

We utilized a methodology to extract Five topics were identified as entities from a dataset comprising 40 research articles focused on climate change adaptation. These keywords represent the core concepts and themes discussed in the articles, forming the foundation for further analysis of their interrelations and significance in the context of adaptation planning. In addition, we selected several five topics specifically related to the urban planning field.

Listing 1: Climate change adaptation

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Adaptive Capacity
Climate-Proofing
Vulnerability Assessment
Ecosystem-Based Adaptation
Resilience
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Listing 2: Urban Planning

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Transportation planning
Zoning
Urban design
Sustainable Development
Land use planning
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Knowledge Graph building method

Method1: Direct Words-Existence Frequency

We employed a straightforward method to calculate words-existence frequency by examining the extent to which the predefined extended seed topics appeared in the plans.

First, we defined the synonyms of the seed topics. For instance, the seed topic "adaptive capacity" includes synonyms such as "adaptive," "adaptation ability,"etc; and "climate proofing" encompasses terms like "climate resilience" and "climate-smart infrastructure," etc. Then to enhance the accuracy of the analysis, we preprocessed the text by removing blank spaces and tokenizing the content. Stop words irrelevant to our topic, such as articles ("a"/"an"), adverbs ("so," "but"), as well as specific terms like "New York City"/"NYC"/"New York," were excluded. Additionally, both the documents and the extended seed topics were lemmatized. These steps were implemented to identify more relevant words and reduce potential biases.

We calculated the weights of the nodes based on the frequency of each topic by dividing the number of occurrences of the topic by the total length of the document. The weights of the edges were determined by constructing a co-occurrence frequency matrix.

Method2: Term Frequency-Inverse Document Frequency (TF-IDF) and Apriori algorithm

To further extract meaningful insights from the planning documents, we employed Term Frequency-Inverse Document Frequency (TF-IDF) as a method for evaluating the significance of Topics. the Apriori algorithm is used to identify frequent patterns of co-occurring Topics in the planning documents, establishing meaningful connections based on their joint appearance.

a) Weighted Entities:

TF-IDF was utilized to assess how critical a word is within the context of a larger corpus of documents, enabling us to compare the overall importance of various keywords in relation to climate change adaptation.

Specifically, for each keyword, a TF-IDF score was calculated. The average score for all keywords within a given topic was then computed to reflect its relative importance across the collection of planning documents.

b) Edges Connection:

The process involves the following steps:

- Transaction Encoder Creation: Each segmented portion of text is treated as a "transaction," forming a matrix of all the keywords found within it. If the document contains the keyword, it was marked as 1; otherwise, it was 0.
- Frequent Itemset Mining: The Apriori algorithm was applied to discover combinations of keywords (itemsets) that met a predefined minimum support threshold (min_support = 0.005), indicating their frequent co-occurrence across documents.
- Association Rule Extraction: Rules are generated to highlight the strength of relationships between entities, using metrics lift (how much the occurrence of one keyword increases the likelihood of another). A minimum threshold (min_threshold = 1) was applied to ensure the relevance of the rules.

$$\mathrm{Support}(A) = \frac{|T(A)|}{|T|}$$

$$\mathrm{Confidence}(A \rightarrow B) = \frac{\mathrm{Support}(A \cup B)}{\mathrm{Support}(A)}$$

$$\mathrm{Lift}(A \rightarrow B) = \frac{\mathrm{Confidence}(A \rightarrow B)}{\mathrm{Support}(B)}$$

Where:

$|T(A)|$: the number of transactions containing itemset A.

$|T|$: the total number of transactions.

Method3: Latent Dirichlet Allocation (LDA) & TF-IDF and Apriori algorithm

This method adopts an indirect approach to build the knowledge graph. First, topic modeling (LDA) is applied to generate the most relevant LDA topics. Next, we compute the co-existence matrix of these LDA topics and use semantic relationships to transfer the weights of the fixed topics of interest. This process allows us to infer the weights of nodes and edges, linking LDA topics to fixed topics, and uncovering indirect frequency of existence and co-existence.

a) Topic Modeling for LDA-topics and Subnodes:

We utilized Latent Dirichlet Allocation (LDA) to identify 10 core LDA-topics. Each topic was further divided into 10 subnodes, some of which are same words, to enhance semantic similarity and improve the effectiveness of the analysis.

b) LDA Subnode Co-existence Calculation:

A co-existence matrix was calculated based on the subnodes of the LDA topics. This matrix quantifies how often LDA subnodes appear together in the same context, providing the basis for subsequent relationship mapping.

c) Semantic similarity between LDA-topics with the fixed-topics.

To align LDA topics with the fixed topics of interest, we calculated semantic similarity. Preprocessing was performed to remove irrelevant words, such as articles (a/an), adverbs (so, but), generic terms (people/professional), and meaningless verbs (improve/decrease). Proper nouns unrelated to the topic, like New York City/NYC/New York, were also excluded. These steps ensured that only meaningful and relevant terms contributed to the topic alignment.

d) indirect fixed-topics Knowledge Graph build up

Finally, we used the weights from LDA subnodes co-existence and semantic similarity to construct the indirect fixed-topics knowledge graph. Nodes represent the fixed topics, while edges indicate indirect relationships inferred through LDA-topics. Edge weights reflect the strength of these connections, enabling the visualization and analysis of implicit topic linkages.

Method4: Direct LLM-model scored frequency

This method leverages large language models (LLMs) to directly evaluate and rank the significance of entities and their relationships. By posing specific questions to the model, we extract scores that quantify the importance of individual entities (nodes) and the strength of connections between them (edges).

a) Entity Value Scoring:

The process begins by asking the LLM to assess the importance of each predefined entity within the context of the planning documents. For example, the prompt might be:

"How important is the term [“climate-proofing”] in the context of climate change adaptation? Assign a score between 0.1 (least important) and 1.0 (most important)."

b) Edge Weight Scoring:

Next, the LLM is asked to evaluate the strength of relationships between pairs of entities. The prompt could be:

"Given the relationship between [“resilience”] and [“climate-proofing”] , how strong is their connection in the given document? Assign a score between 0.1 (weakest connection) and 1.0 (strongest connection)."

Knowledge Graph construction using the LLM-derived scores. This direct evaluation bypasses the need for intermediate statistical calculations.

Method5: Combined LLM and Embedding-based method

This method proposes a combined approach to integrate embedded nodes with manually defined nodes, creating a unified representation for knowledge graph construction.

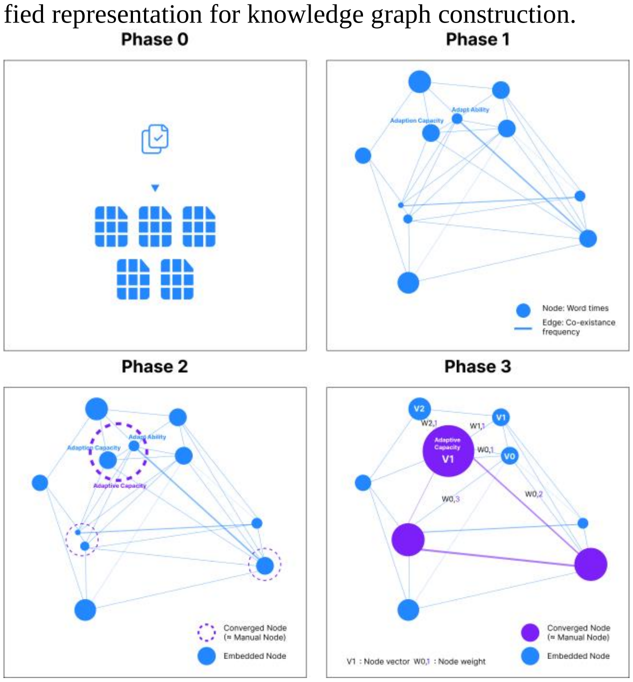

Figure 1: Process for Combined LLM and Embedding-based method.

a) Phase 0: File Chunking

Load master planning files and split them into manageable chunks. Each paragraph is divided based on a defined chunk_size and chunk_overlap, ensuring text continuity while maintaining computational efficiency.

b) Phase 1: Embedded Nodes and Edge Generation

Using an LLM, summarize the key nodes (concepts) within each paragraph. Count their occurrences and analyze the frequency of co-occurrence between nodes.

Node Representation: Larger nodes represent higher importance (based on frequency).

Edge Representation: Thicker edges indicate stronger relationships (based on co-occurrence).

c) Phase 2: Converged Nodes

Iteratively compute semantic similarity between manually defined nodes and embedded nodes. When the semantic similarity reaches or exceeds a threshold of 95%, the embedded nodes are merged into converged nodes, representing a refined alignment with manual definitions.

Purpose: Semantic similarity thresholds ensure conceptually aligned terms are merged, minimizing noise while enhancing the graph's accuracy.

d) Phase 3: Node Vector Calculation

For each converged node, calculate its vector representation based on its connections to embedded nodes. The relationship between a converged node and its linked embedded nodes is expressed as:

$$\mathrm{V_p} = \mathrm{W_{0,p}} \times \mathrm{V_0} + \mathrm{W_{1,p}} \times \mathrm{V_1} + \mathrm{W_{2,p}} \times \mathrm{V_2} + \cdots + \mathrm{W_{n,p}} \times \mathrm{V}_n$$

Where:

$\mathrm{V_p}$: The vector of the converged node p.

V_n : The vector of the nth connected embedded node.

$\mathrm{W_{n,p}}$: The weight (relationship strength) between the converged node p and the embedded node n.

Knowledge Integration and Visualization

To visualize the extracted knowledge, we constructed a domain knowledge graph, integrating weighted keywords and their associations.

Tools NetworkX were used to build the graph. And different Weight comparison is important for this study. While the node size and color indicate the significance of a keyword, with larger nodes representing higher weights and two type of nodes are in different color; And Edge thickness reflects the strength of relationships, where thicker edges signify stronger associations between keywords.

Validation: Expert

To validate the knowledge graph, domain experts review and rank the graphs based on their professional judgment. They assess the importance of specific nodes, noting cases where certain nodes are less critical than suggested by the methods. Additionally, experts evaluate the strength of connections between nodes to ensure the relationships align with their expertise.

Result

Comparison of methods 1,2,3,4 between New York and Shanghai

This section presents the evaluation metrics and methods used to assess the performance and reliability of the knowledge graph and associated tools developed during this study.

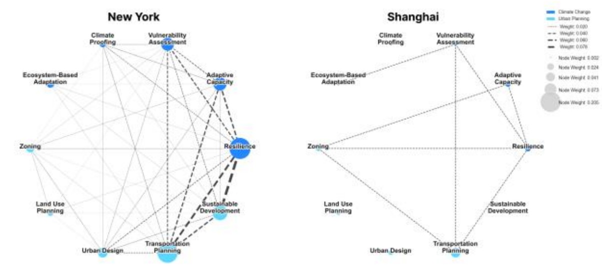

Figure 2: KG for method1
direct words-existence frequency

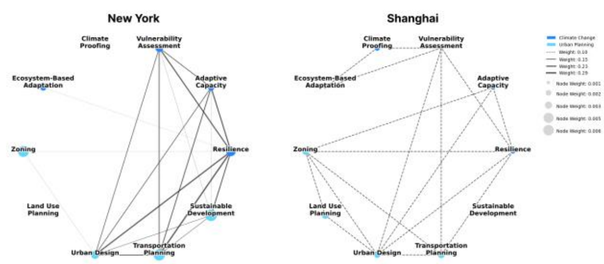

Figure 3: KG for method2
TF-IDF and Apriori algorithm frequency

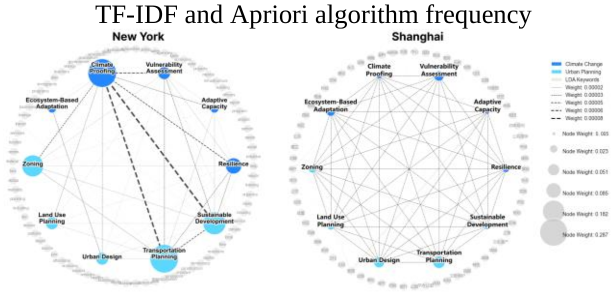

Figure 4: KG for method3
LDA & TF-IDF model and indirect co-existence frequency

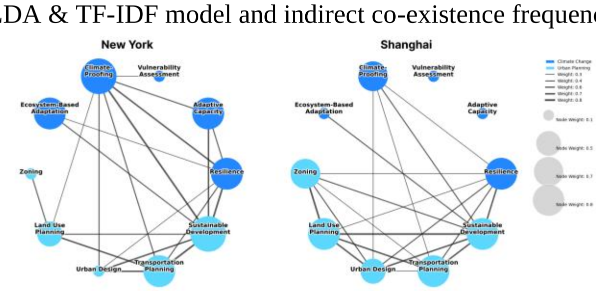

Figure 5: KG for method4
direct LLM-model scored frequency

Probable Comparison and Evaluation

This subsection evaluates the process of generating keywords for the knowledge graph, comparing the results between different datasets (e.g., New York vs. Shanghai) and different method to analyze their completeness, accuracy, and alignment with the respective planning contexts. Metrics include:

- **Precision**: How relevant are the generated keywords to the climate adaptation domain?
- **Recall**: How comprehensive is the keyword extraction in covering critical concepts?
- **Divergence**: Analysis of differences in keyword emphasis and representation across datasets.

By comparing commonalities and differences across multiple knowledge graphs, insights can be gained into the suitability of various methods for knowledge graph analysis in urban planning. This process helps identify which approaches are most effective in capturing domain-relevant knowledge graphs.

Limitation and Constrains

Current expert evaluation is affected by the expert's knowledge limitation, and there should be more experience expert who has the experience of the master plan domain, as well as the specific climate adaptation background. Besides, the method of the direct LLM model score is more comparable for different master plan, whereas it has the constrain of hallucination. Therefore, some implementation of the combined way may be further explored.

Conclusion

This study applies different methods for knowledge graph generation from urban master plans in New York City and Shanghai to understand their approaches to climate change adaptation and urban planning. Five main aspects of great concern in climate change adaptation and five in urban planning were identified first as the nodes of the knowledge graph. Then, different techniques were applied to the two master plans to compare traditional information retrieval and new large language model (LLM) methods in generating edges connecting those concept nodes. The KGs generated will be validated against human evaluation. The results suggest that while the direct LLM method can achieve better semantic results, it lacks explainability critical for practice. A new LLM embedding-based method is proposed to combine domain expert knowledge, traditional explainable methods, and LLM embeddings to provide a more interpretable and accurate representation of concepts as KGs. The KGs generated from the two cities with different contexts also suggested the impact of different planning systems on how master plans are organized and written. The findings will be helpful to generate KGs to represent concepts and their relationships in master plans, which embed various planning theories and practice consideration in different social, economic, and regulatory contexts. The comparison will provide the basis for planning knowledge and policy transfer across different cities and cultures. This methodology offers a scalable approach for analyzing planning documents across different cities, providing valuable insights for global urban adaptation strategies. Future research could extend this framework to additional cities or incorporate dynamic changes over time to better understand the evolution of urban adaptation planning.

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